

Collaborative AI Travel Planning Platform

Project Proposal and Business Case

Document Information

Author(s):	Project team
Date:	August 1, 2026
Keywords:	capstone, AI, travel planning, collaborative decision making, business case

Document Revision History

Version	Date	Author	Changes Made
1.0	August 1, 2026	Project team	Initial proposal draft for new capstone direction

Reviewers

Name	Contact Information
Instructor / team reviewer	To be confirmed

Introduction

This project proposes a collaborative AI travel planning platform that helps small groups make travel decisions together without forcing one person to manually collect preferences, compare constraints, and negotiate compromises. The product is not a generic itinerary generator. Its main value is helping a group reach aligned decisions on destination, dates, budget, lodging, activities, and local adjustments while keeping each member's preferences, hard constraints, and privacy boundaries visible and manageable.

The initial capstone version focuses on a practical MVP that is feasible within a short academic delivery timeline. The MVP will support complete-trip planning for small friend groups and small company outings, while keeping advanced booking, payment, and real-time disruption handling out of scope.

Business Problem and/or Opportunity

Group travel planning is often inefficient, frustrating, and unfair. A single organizer usually has to gather scattered input from chat messages, remember each person's budget and availability, compare conflicting preferences, and repeatedly rebuild plans when one condition changes.

This creates several problems: decision fatigue slows planning, important constraints are easy to miss, louder participants can dominate choices, and most existing travel tools optimize for individuals rather than group consensus.

This creates an opportunity for a collaborative planning platform that acts as a structured decision engine for group travel. Instead of only suggesting destinations, the system can collect preferences, separate hard constraints from soft preferences, calculate compromise options, explain why a recommendation was chosen, and help groups converge on an acceptable shared plan faster.

Project Outcomes

The project will deliver a web-based prototype that demonstrates how AI can support group travel decision making from trip setup to proposal comparison and iterative revision.

The expected outcomes are listed below:

Success Criteria (including measurement metrics)

The capstone project will be considered successful if the prototype can demonstrate a complete planning cycle, generate at least one valid recommendation that respects hard constraints, and support revision without overwriting already accepted trip sections.

Suggested evaluation metrics are listed below.

Scope

The MVP focuses on the core collaborative decision loop and enough supporting features to demonstrate real product value within the capstone timeline.

Out of Scope

To keep the capstone deliverable achievable, several advanced or integration-heavy capabilities are intentionally excluded from the MVP.

End State

At the end of the capstone, the team will deliver a working prototype that demonstrates the full core planning loop for a small group trip: create a trip, gather structured inputs, generate candidate

plans, review tradeoffs, record feedback, and revise affected plan sections when requirements change.

Assumptions

The proposal assumes that users are willing to provide structured input, that mock or curated travel data can be used for the MVP, and that the course deliverable values business relevance and feasibility over production-scale integrations.

Known Risks

The main risks are scope growth, weak AI explanation quality, poor input structure, limited testing time, and revision logic that could unintentionally overwrite previously accepted decisions.

Appendices

The appendices summarize product positioning, MVP design principles, and future release ideas.

- A structured group trip workspace where members submit budgets, availability, preferences, and non-negotiable constraints.
- An AI-assisted recommendation flow that produces one or more candidate trip plans based on group input.
- A transparent explanation layer that shows tradeoffs, compromise logic, and satisfaction impact.
- A collaboration workflow that records approvals, feedback, and revision requests without overwriting already accepted sections.
- Separate experiences for end-users and admins so that platform operations and trip management can be supervised.
- Time to first complete proposal
- Percentage of hard constraints satisfied
- Number of revision cycles required before group acceptance
- Member satisfaction score for the proposed plan
- Percentage of proposal sections preserved after a change request

In-Scope Product Features

- Trip creation and group invitation
- Structured preference collection for destination, dates, budget range, accommodation needs, activity interests, dietary or accessibility constraints, and other travel conditions
- Separation of hard constraints and soft preferences
- AI-generated candidate trip proposals with rationale
- Group review, feedback, and lightweight approval tracking
- Controlled revision flow when a member changes a key condition

- Visibility into member satisfaction and unresolved conflicts
- Admin tools to monitor trip records, user activity, and planning status

End-User Functionality

- Create or join a trip workspace
- Submit personal constraints and preferences
- Review AI-generated proposals
- Compare options and tradeoffs
- Accept, reject, or request changes to specific sections
- View current group alignment and pending decisions

Admin Functionality

- View all trip workspaces and their statuses
- View user participation and planning progress
- Manage trip records and platform-level settings
- Review issue flags or stalled planning flows
- Export or summarize trip planning activity for reporting

Out-of-Scope Items

- Real flight and hotel booking integration
- Payment processing
- Multi-city itinerary optimization
- Coordination for travelers departing from different origin cities
- Automatic replanning for delays, cancellations, or live disruptions
- Public social matching with strangers
- Complex visual version diffing across full trip plans
- Production-grade commission or subscription billing flows

Assumptions Detail

- Users are willing to provide structured input instead of relying only on free-form chat.
- The capstone team can use mock or curated travel data for the MVP.
- A lightweight AI workflow is sufficient to demonstrate compromise recommendations and explanation quality.
- Privacy expectations can be addressed in the prototype through selective visibility of member inputs and constraints.

Known Risks Detail

- Scope growth could make the project too large for the capstone timeline.
- AI-generated suggestions may be perceived as inaccurate or unfair if explanations are weak.
- Poorly structured input collection could reduce recommendation quality.
- If change-handling logic is unclear, revised plans may overwrite previously accepted decisions and reduce user trust.

Appendix A: Core Product Positioning

- The product should be framed as a complete-trip collaborative planning platform rather than only a local recommendation tool. Sub-decisions such as where to eat, where to stay, what to do, or how to adjust a route should support the larger trip workflow rather than define the product by themselves.

Appendix B: MVP Principle Summary

- Respect hard constraints before optimizing preferences.
- Keep member consent explicit.
- Separate satisfaction scoring from acceptance state.
- Preserve accepted sections when revising a plan.
- Explain compromise logic transparently.
- Prioritize one-month MVP feasibility over feature breadth.

Appendix C: Future Release Ideas

- Booking and payment integration
- Smarter local re-planning during the trip
- Family or advanced shared budget management
- Richer role and permission controls
- Business-to-business trip coordination features

Target Audience

End-User Description

Primary end-users are friend groups planning leisure travel together and small company teams planning outings or retreats. These users need a simple interface that reduces coordination overhead, protects individual constraints, and helps the group reach a practical decision without endless chat back-and-forth.

Admin User Description

Admin users may include platform operators, business owners, internal coordinators, or team organizers with elevated management permissions. These users need visibility into trip workspaces, user participation, planning progress, and basic operational records.

User Interface Design

The end-user interface will prioritize clarity, structured forms, transparent recommendation explanations, and section-based review. The admin interface will prioritize visibility, status tracking, and operational summaries.

Technical Details

This section provides a planning-level technical direction only and does not commit the team to final implementation choices before deeper validation.

Technology Stack

- Front-End: React, HTML5, CSS
- Back-End: Python with Flask or a comparable lightweight API framework
- Database: PostgreSQL
- AI Layer: LLM-based recommendation and explanation services

Architecture Overview

At a high level, the front-end will collect user inputs and display generated proposals, the back-end will manage business rules and workflow state, the database will store users, trips, preferences, decisions, and revisions, and the AI layer will transform structured inputs into explainable recommendations.